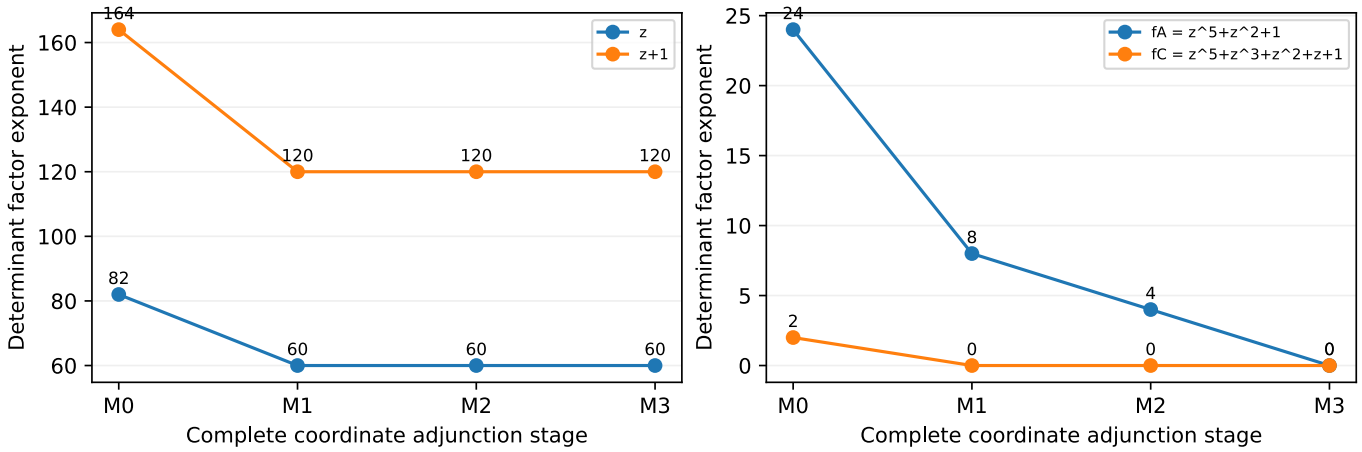
R53: exact defect supports and marked correspondence coordinates

Complete source-bound trace geometry

All 32 generator traces and all 1, 024 Gram entries agree with an independent companion-power and quadratic-base trace calculation. Both full source and discriminant-change products pass; changed entries are rejected. All eight predictions match in discrim53-02, with

```
fA=z^5+z^2+1; fC=z^5+z^3+z^2+z+1
```

```
disc(B)=z^76(z+1)^24 fC^12; Norm(Delta)=z^11(z+1)^22 fC
```



Measured complete determinant exponents, not a sample or a timing comparison. Source: runs/discrim53-02/native.log. The last two adjunctions remove defect only at fA. Final coordinate defect remains at z=0, 1. Completed local-ring identification needs branch valuations and residue-field descent.

Actual denominator and marked model

The composed membership gives the full witness

```
(G,N,psi3,alpha)*(Vx,Vy,U,Vz0)^t = Delta^16
```

Alpha is the actual R52 norm coefficient, and the G, N, psi3 quotient has dimension 64. The complete witness is retained in raw custody. Generic coefficient denominators and exceptional fibers remain explicit obligations.

Exact source geometry to the normalized correspondence

Trace pairing

32 generator traces; 1024 entries

All eight discriminant predictions match

Next: completed rings at z=0,1

Actual image-line denominator

Full witness (G,N,psi3,alpha) W = Delta^16

Nonzero quotient dimension 64

Generic registered chart

Marked quotient normalization

Four explicit unit products

Full curve change and corruption check

Next: target QR / F2 map / 11 action

Full integral fiber versus the localized compact equations

At c=1 (A=0), the source equations give

```
Q=lambda^4+b*lambda+b^2; G=Q^4; N=b^3*Q^2
```

The prediction N=0 failed. A fresh native check verifies both full ideal products identifying (G, N) with (Q^2). This compact specialization leaves nu free: its two-variable quotient has infinite dimension (vdim=-1).

The full three QR equations instead give a finite etale algebra of dimension 16, with a Jacobian-minor unit witness and complete ideal products. This holds over F2(b) and at the actual F32 base point

```
a^5+a^2+1=0; V=(1+a^4+a)/(a+a^2); c=1; b=1+1/a
```

The original base equation and b, c formulas are checked directly. The lambda quartic factors with degrees 1, 1, 2 over F32, so it has four distinct geometric roots, two of which are F32 rational.

The actual twisted fiber

Ordinary nu is descended nu+b*zeta at A=0, where zeta^2+zeta+1=0. Elimination and both full augmented-ideal products certify descent. The descended quotient has dimension 16, Jacobian unit, and leading monomials lambda^4 and nu^4. Its standard monomials are lambda^i*nu^j for i, j = 0, 1, 2, 3. Thus nu separates four points over each geometric lambda root. The ordinary and twisted nu equations differ; both are retained, without identifying them.

This identifies the missing integral coordinate data at the projection fiber. It is a finite-fiber result; direct binding to the normal RR fiber, the remaining completed local rings and proper correspondence are next. Sources: supersingular53-01 and compactfiber53-01, including every full witness.

Normalized map prefix and its cap

packet53-02 retains the composed inverse, the image line, normalized coordinates and a native target-Delta unit check. The coordinate export is 78 KiB; the Delta unit lift was not separately serialized. Direct multivariate modular powers reached the original 1200 second cap(exit124) before the five target equations completed. This is a computational cap, not a failed identity. The next checker can use Frobenius squares and the retained QR/G, N syzygies, or the actual degree-8 level-11 base to reduce coefficient growth.

Validation and next construction

The generic marked Tate change uses h=A*r+b, s=(r^2+a4+A*q)/h, D2=b+r+s^2+A*s and u=h/D2. All t, h, D2, u units require full Bezout products. The complete curve change is oldX=u^2*x+r and oldY=u^3*y+s*u^2*x+q, with bnew=D2/u^2=h/u^3 and Anew=A/u. See CM.md for actual terminal acceptance.

Complete terminal attempt ledger

Attempt	Result	Native / Lean receipt
alphaunit53-01	accepted	exit 0; 0 error lines
compactfiber53-01	accepted	exit 0; 0 error lines
discrim53-01	rejected	exit 0; 89 error lines
discrim53-02	accepted	exit 0; 0 error lines
packet53-01	rejected	interrupted; intermediate custody invalid
packet53-02	rejected	exit 124; 0 error lines
supersingular53-01	accepted	exit 0; 0 error lines
tate53-01	rejected	exit 0; 106 error lines
tate53-02	rejected	exit 0; 8 error lines
tate53-03	accepted	exit 0; 0 error lines
transfer-lean53-01	rejected	compile 0; audit 1
transfer-lean53-02	accepted	compile 0; audit 0

Every failed implementation attempt is retained, including late markers after errors. Exact native limits: seed 53, one thread, 4 GiB virtual, 1200 s. No accepted producer rerun for publication. Lean uses 4.34.1, pinned mathlib d13f23b, -j1, -M4096, 1200s separately for compile/audit, without a virtual-memory cap. Native wall times are recorded in results.json; complete CPU times were not recorded. These are exact identity checks, not statistical or performance comparisons.

packet53-01 has invalid intermediate custody: a live Bash-runner edit caused interpreter re-entry into a second native invocation. The owned processes were stopped, all retained files are rejected, and this exception is excluded from valid historical executed-byte pins. A fresh workflow requires completed Lean receipts before admission. All 12workflows are terminal, 6 accepted and 6 rejected; 24 valid completed-workflow input pins are verified separately.

The three Lean implications cover determinant change, composed alpha product and the resulting unit. Actual arrays, trace semantics, source specialization and geometric hypotheses are external inputs to those statements.

Next decisive checks

Use the normalized line coordinates (L+s)/u and (L*r+N+q)/u^3. Verify all target QR identities, the F2 descent and registered denominator specializations. Then specialize the level11 base and certify intended subgroup action. The two completed local rings, proper polarization, full quotient saturation and arbitrary reverse-divisor evaluation retain their separate obligations.

Sources: R50 complete normal RR algebra; R51 line/collision/quotient functions; R52 complete nu closure and norm line; pinned donors.json. Established isogeny model conventions: Sage elliptic-curve isogeny documentation. Proper/function-field correspondence: Stacks Project tag0BXX. No novelty claim is made for trace pairings, Bezout composition or Tate normalization.